

**Course: Fundamentals of Generative AI and Large Language
Models: Theory and Practice**

**Week 3: Variational Autoencoders (VAEs):
Foundations**

Week 3 Outline

- ✓ Intuition behind KL divergence
- ✓ VAE architecture
 - ✓ Encoder
 - ✓ Latent-space sampling
 - ✓ Decoder
- ✓ Evidence Lower Bound (ELBO)
- ✓ VAE Loss Function
 - ✓ Reconstruction loss
 - ✓ KL – divergence loss
- ✓ Reparameterization trick
- ✓ Numerical example illustrating the complete working of a VAE

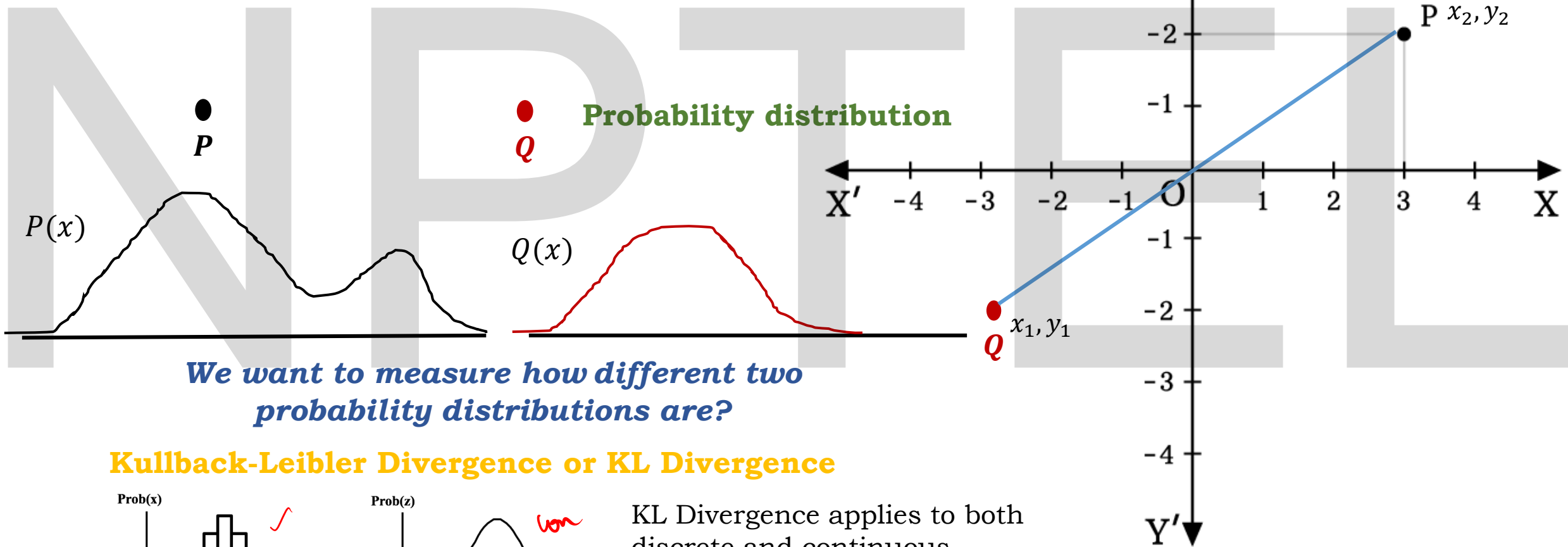
Hands-on : There will be no hands-on session during Week 3. The hands-on implementation of the VAE will be covered in Week 4.

Today's Session Overview

- Intuition behind KL divergence
 - ✓ Basic concept of KL divergence
 - ✓ Forward and reverse KL divergence
 - ✓ Mode-covering and mode-seeking behaviour

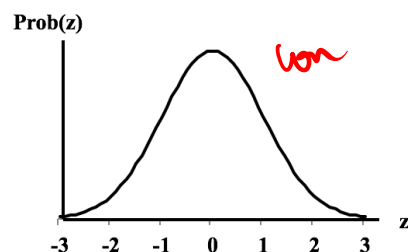
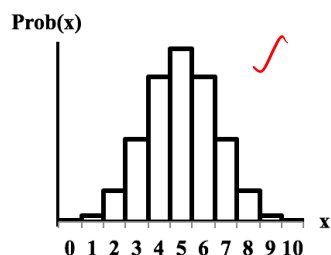
Imagine we have two points P and Q in the 2D space. Then measuring the distance between them is simple.

$$\text{Euclidean Distance} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$



We want to measure how different two probability distributions are?

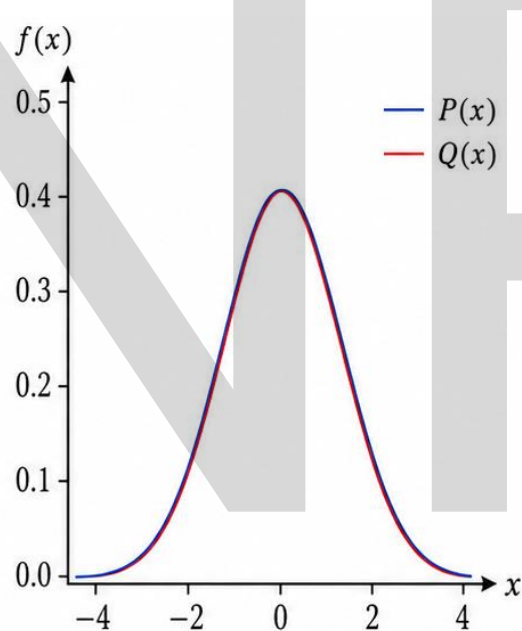
Kullback-Leibler Divergence or KL Divergence



KL Divergence applies to both discrete and continuous probability distribution.

What is KL Divergence?

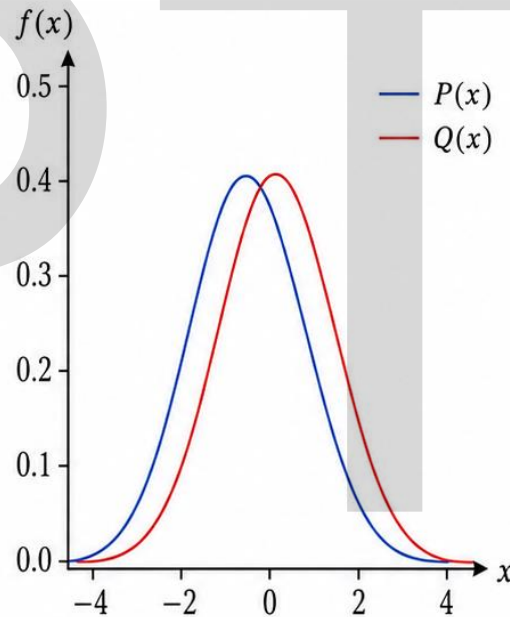
KL divergence measures how much a probability distribution $Q(x)$ differs from a reference probability distribution $P(x)$.



Identical distributions

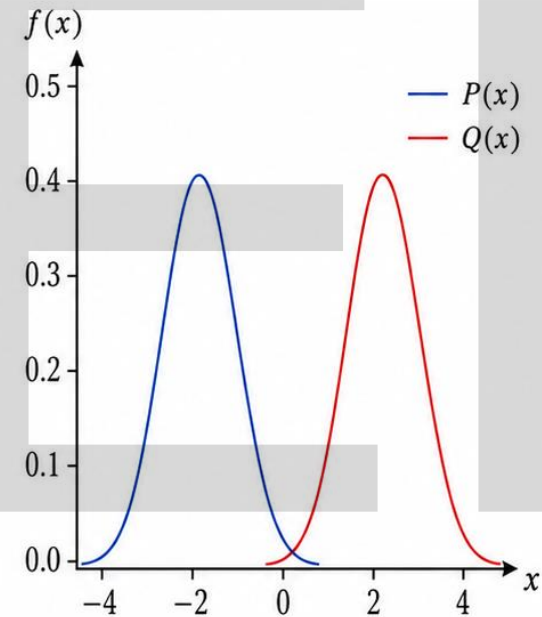
$$\therefore P(x) = Q(x).$$

The two distributions completely overlap.
The KL divergence is zero.



Slightly different distributions:

The distributions have nearby means and good overlap.
The KL divergence is small.



Very different distributions:

The distributions have widely separated means and very little overlap.
The KL divergence large.

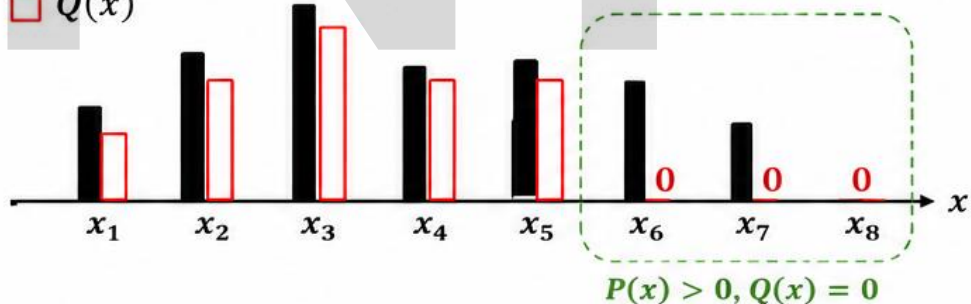
Consider two probability distributions $P(x)$ and $Q(x)$ then:

KL Divergence in two directions

Forward KL (Discrete): Strongly penalizes Q when it fails to cover regions where P has probability mass.

$$D_{KL}(P||Q) = \sum_x P(x) \log \frac{P(x)}{Q(x)}$$

■ $P(x)$
□ $Q(x)$

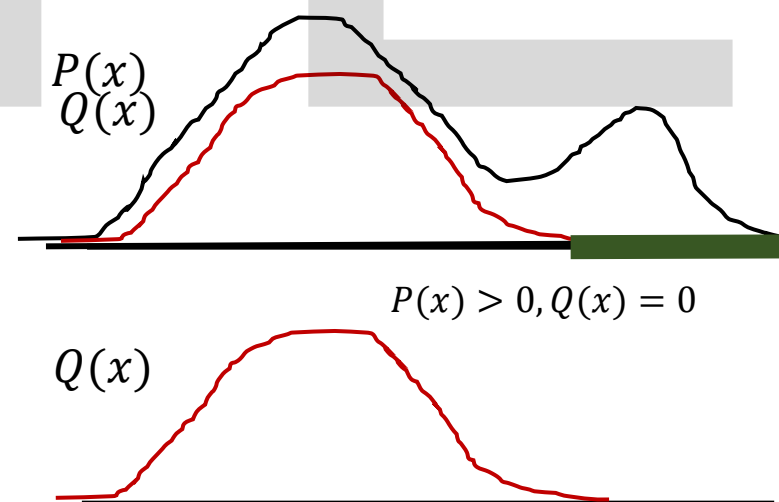


Since $P(x) > 0$ and $Q(x) = 0$, then $D_{KL}(P || Q)$ becomes infinite because $\log \frac{P(x)}{0} \rightarrow \infty$.

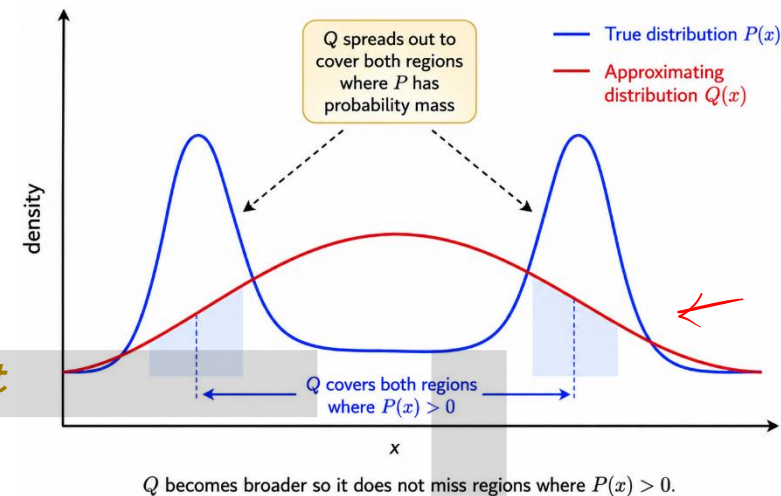
So forward KL pushes Q to spread out and cover the complete support of P .

Continuous Probability Distribution

$$D_{KL}(P || Q) = \int P(x) \log \frac{P(x)}{Q(x)} dx$$



Forward KL: mode-covering behavior



Consider two probability distributions $P(x)$ and $Q(x)$ then:

KL Divergence in two directions

Continuous Probability Distribution

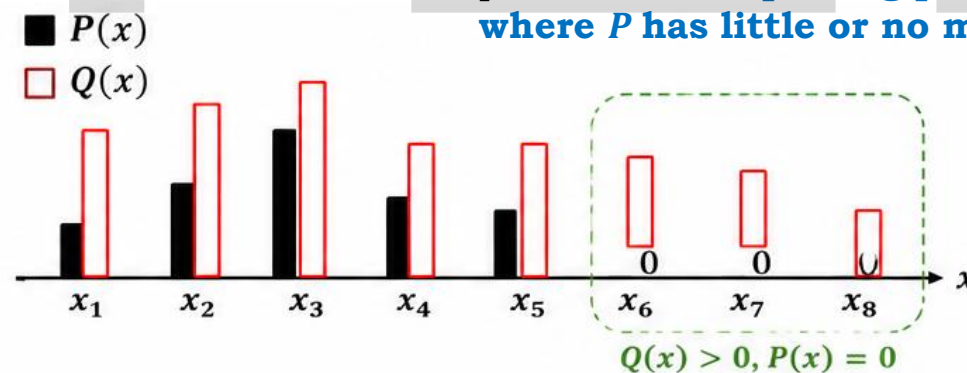
$$D_{KL}(Q \parallel P) = \int Q(x) \log \frac{Q(x)}{P(x)} dx$$

If $Q(x) > 0$ in a region where $P(x) \approx 0$ or is extremely small, then $\frac{Q(x)}{P(x)}$ becomes extremely large. Therefore, the reverse KL penalty becomes very large.

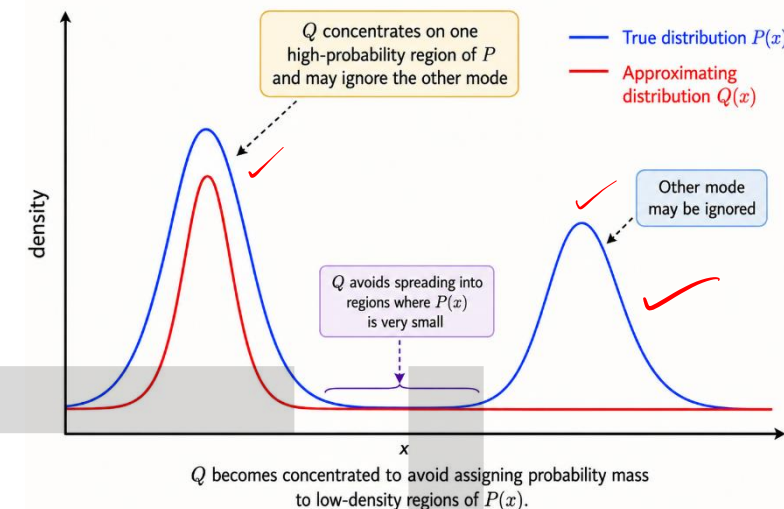
Reverse KL: If the learned distribution $Q(x)$ spreads into regions where the true distribution $P(x)$ has no mass or very little mass, reverse KL penalizes it.

$$D_{KL}(Q \parallel P) = \sum_x Q(x) \log \frac{Q(x)}{P(x)}$$

penalizes **Q for placing probability where P has little or no mass.**



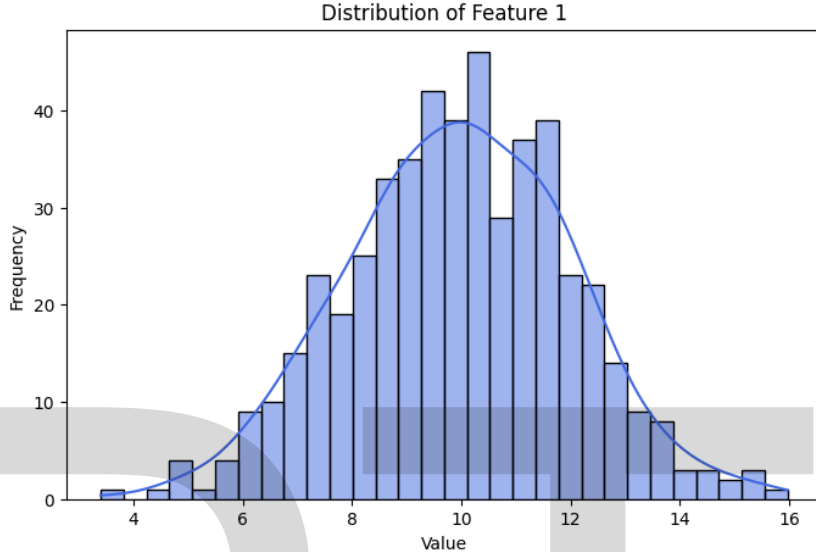
Reverse KL: mode-seeking behavior



Sample dataset

Feature 1	Feature 2
10.99342831	14.77853264
9.723471398	17.72824992
11.29537708	7.804297279
13.04605971	13.68890771
9.531693251	10.04807229
9.531726086	10.53862385
13.15842563	10.22281823
11.53486946	9.408027691
9.061051228	12.14556488
11.08512009	9.507149651
9.073164614	12.81137048
9.068540493	11.84928567
10.48392454	11.28315586
6.173439511	9.277309014
6.550164335	10.26968601
8.875424942	14.26617368
7.974337759	13.50275156
10.62849467	9.067334266
8.183951849	12.29799692

$P(x)$



Real dataset samples are drawn from an unknown true distribution:

$P(x)$

A generative model learns an approximate distribution:

$Q_{\theta}(x)$

Goal:

$$Q_{\theta}(x) \approx P(x)$$

KL divergence quantifies the mismatch between the two distributions.

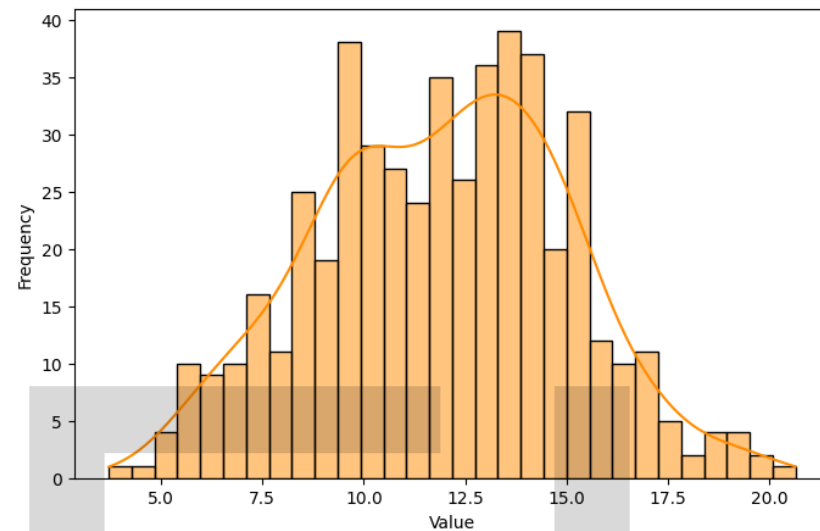
For Forward KL:

$$D_{KL}(P(x) \parallel Q_{\theta})$$

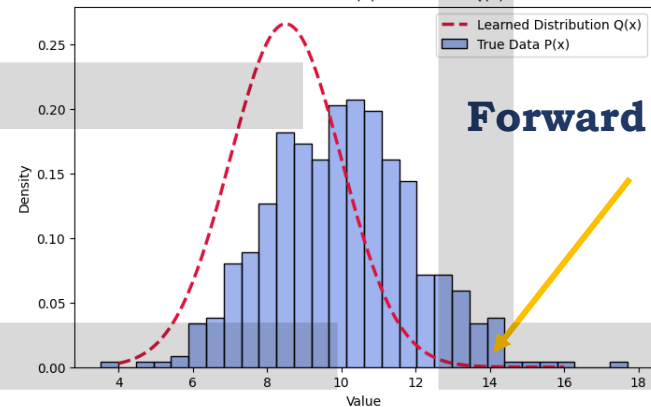
For Reverse KL:

$$D_{KL}(Q_{\theta} \parallel P(x))$$

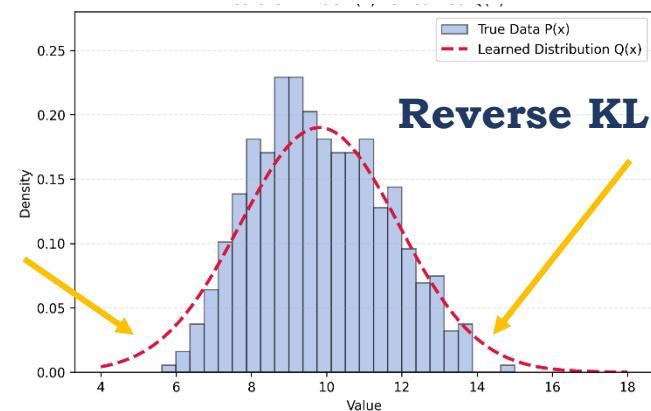
Distribution of Feature 2



Feature 1: True P(x) vs Learned Q(x)



Forward KL



Reverse KL

Suppose the real data contain:

- short people around 160 cm,
 - tall people around 190 cm,
- with very few people around 175 cm.

The true distribution $P(x)$ has two modes.

Forward KL vs. Reverse KL: Mode-Covering and Mode-Seeking

Cluster A and *Cluster B*

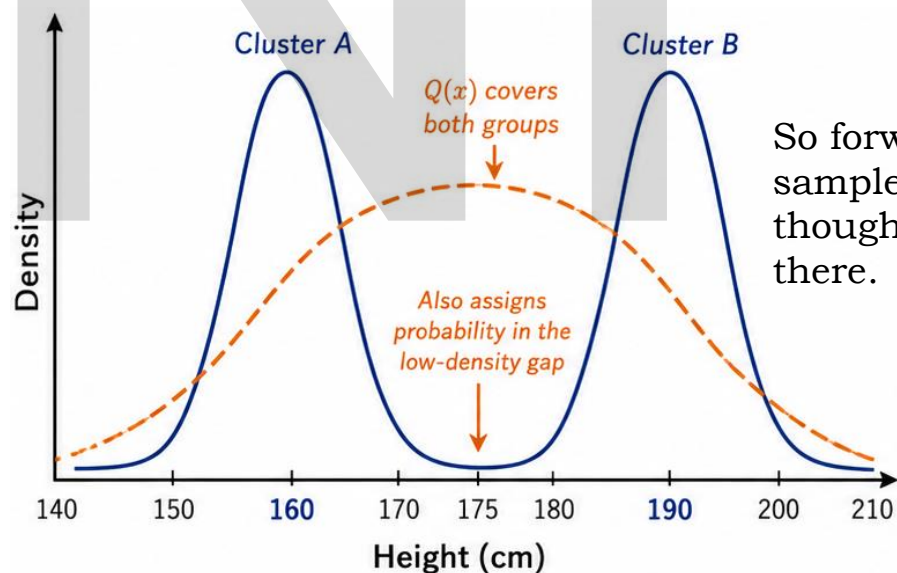
and very little data exists between them.

Forward KL: Mode-covering $D_{KL}(P||Q)$

A single Gaussian $Q(x)$ becomes broad to cover both modes of $P(x)$.

Advantage: Both groups are represented.

Limitation: $Q(x)$ also assigns probability to the low-density region around 175 cm.



So forward KL may generate samples around 175 cm, even though very few real people occur there.

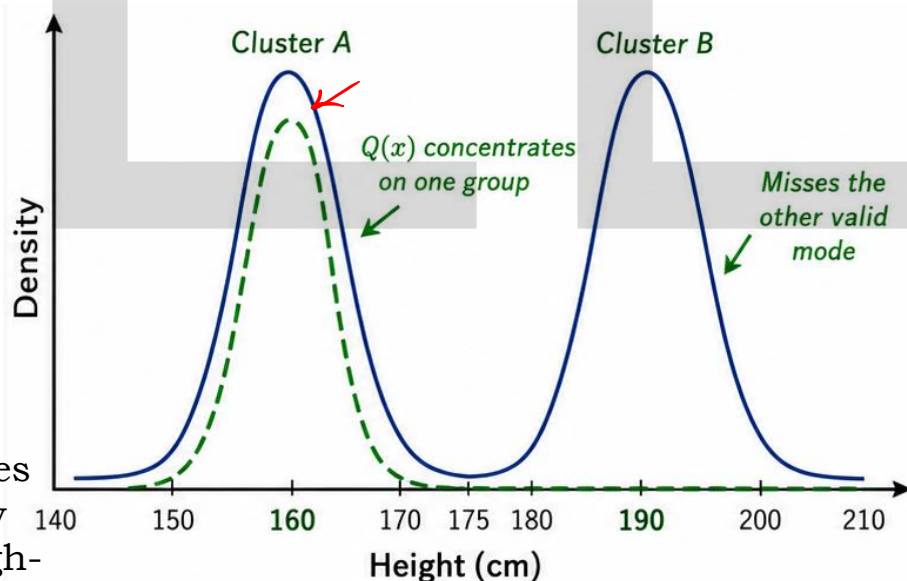
Reverse KL ignores the other valid group. But the samples it generates from the chosen group are usually realistic and concentrated in a high-density region.

Reverse KL: Mode-seeking $D_{KL}(Q||P)$

A single Gaussian $Q(x)$ may concentrate on either the 160 cm mode or the 190 cm mode.

Advantage: Samples from the selected mode are realistic.

Limitation: The other valid mode may be missed.



Consider a dataset containing cats and dogs images.

Task: A model has to generate animal images from this dataset

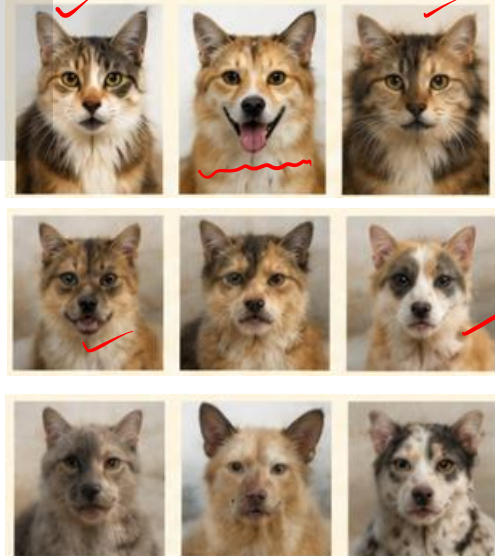
Forward KL

The model tries to cover both cats and dogs.

If its model capacity is limited, it may produce ambiguous samples that look partly like a cat and partly like a dog.

Forward KL (mode covering): Covers all modes but may generate ambiguous in-between samples.

Forward KL (mode covering)
with limited capacity



Reverse KL

The model may learn only cats or only dogs and generate clear cat or dog images, but fail to generate other class.

Reverse KL (mode seeking): Generates sharp, realistic samples from one mode but misses other valid modes.

Reverse KL (mode seeking)
with limited capacity

Only one mode is captured.



(Dogs are missed)

Only one mode is captured.



(Cats are missed)

Only one mode is captured.



(Dogs are missed)

Next Session

- Self-information, entropy, cross-entropy, and the detailed derivation of KL divergence

NPTREL

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Thank you